

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 1 of 16

#### SECTION 1: Identification of the substance/mixture and of the company/undertaking

##### 1.1. Product identifier

ARC BX2(E) Part A, ARC BX5(E) Part A

##### 1.2. Relevant identified uses of the substance or mixture and uses advised against

###### Uses advised against

No information available.

##### 1.3. Details of the supplier of the safety data sheet

|                          |                               |                               |
|--------------------------|-------------------------------|-------------------------------|
| Company name:            | Chesterton International GmbH |                               |
| Street:                  | Am Lenzenfleck 23             |                               |
| Place:                   | D-85737 Ismaning GERMANY      |                               |
| Telephone:               | +49 89 99 65 46 - 0           | Telefax: +49 89 99 65 46 - 50 |
| e-mail:                  | eu-sds@chesterton.com         |                               |
| e-mail (Contact person): | eu-sds@chesterton.com         |                               |
| Internet:                | www.chesterton.com            |                               |
| Responsible Department:  | eu-sds@chesterton.com         |                               |

##### 1.4. Emergency telephone number:

+49(0) 551 - 1 92 40 (GIZ-Nord, 24h)

#### SECTION 2: Hazards identification

##### 2.1. Classification of the substance or mixture

###### Regulation (EC) No 1272/2008

Skin Irrit. 2; H315  
Eye Irrit. 2; H319  
Skin Sens. 1; H317  
Aquatic Chronic 3; H412

Full text of hazard statements: see SECTION 16.

##### 2.2. Label elements

###### Regulation (EC) No 1272/2008

###### Hazard components for labelling

2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran  
Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl}oxirane

Signal word: Warning

Pictograms:



###### Hazard statements

|      |                                      |
|------|--------------------------------------|
| H315 | Causes skin irritation.              |
| H317 | May cause an allergic skin reaction. |

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 2 of 16

H319 Causes serious eye irritation.  
 H412 Harmful to aquatic life with long lasting effects.

#### Precautionary statements

P264 Wash hands thoroughly after handling.  
 P273 Avoid release to the environment.  
 P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.  
 P302+P352 IF ON SKIN: Wash with plenty of water.  
 P333+P313 If skin irritation or rash occurs: Get medical advice/attention.  
 P305+P351+P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.  
 P337+P313 If eye irritation persists: Get medical advice/attention.  
 P362+P364 Take off contaminated clothing and wash it before reuse.

#### 2.3. Other hazards

The safety and health hazards are detailed separately for Part A and Part B. The final cured material is considered nonhazardous. Upon machining, refer to the precautions in the safety data sheets for Part A and Part B.

### SECTION 3: Composition/information on ingredients

#### 3.2. Mixtures

##### Hazardous components

| CAS No     | Chemical name                                                                                                                                                                                             |              |                  | Quantity    |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|-------------|
|            | EC No                                                                                                                                                                                                     | Index No     | REACH No         |             |
|            | Classification (Regulation (EC) No 1272/2008)                                                                                                                                                             |              |                  |             |
| 1675-54-3  | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran                                                                                                                                         |              |                  | 15 - < 20 % |
|            | 216-823-5                                                                                                                                                                                                 | 603-073-00-2 | 01-2119456619-26 |             |
|            | Skin Irrit. 2, Eye Irrit. 2, Skin Sens. 1, Aquatic Chronic 2; H315 H319 H317 H411                                                                                                                         |              |                  |             |
|            | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl}oxirane |              |                  | 5 - < 10 %  |
|            | 701-263-0                                                                                                                                                                                                 |              | 01-2119454392-40 |             |
|            | Skin Irrit. 2, Skin Sens. 1, Aquatic Chronic 2; H315 H317 H411                                                                                                                                            |              |                  |             |
| 100-51-6   | benzyl alcohol                                                                                                                                                                                            |              |                  | 1 - < 5 %   |
|            | 202-859-9                                                                                                                                                                                                 | 603-057-00-5 | 01-2119492630-38 |             |
|            | Acute Tox. 4, Acute Tox. 4, Eye Irrit. 2; H332 H302 H319                                                                                                                                                  |              |                  |             |
| 13463-67-7 | titanium dioxide                                                                                                                                                                                          |              |                  | < 1 %       |
|            | 236-675-5                                                                                                                                                                                                 | 022-006-00-2 | 01-2119489379-17 |             |
|            | Carc. 2; H351                                                                                                                                                                                             |              |                  |             |

Full text of H and EUH statements: see section 16.

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 3 of 16

#### Specific Conc. Limits, M-factors and ATE

| CAS No     | EC No     | Chemical name                                                                                                                                                                                             | Quantity    |
|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|            |           | Specific Conc. Limits, M-factors and ATE                                                                                                                                                                  |             |
| 1675-54-3  | 216-823-5 | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran                                                                                                                                         | 15 - < 20 % |
|            |           | inhalation: LC50 = ca. 24,6 mg/l (vapours); dermal: LD50 = > 2000 mg/kg; oral: LD50 = 19800 mg/kg Skin Irrit. 2; H315: >= 5 - 100 Eye Irrit. 2; H319: >= 5 - 100                                          |             |
|            | 701-263-0 | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl}oxirane | 5 - < 10 %  |
|            |           | dermal: LD50 = > 2000 mg/kg; oral: LD50 = > 5000 mg/kg                                                                                                                                                    |             |
| 100-51-6   | 202-859-9 | benzyl alcohol                                                                                                                                                                                            | 1 - < 5 %   |
|            |           | inhalation: ATE = 11 mg/l (vapours); inhalation: LC50 = >4,178 mg/l (dusts or mists); dermal: LD50 = > 2000 mg/kg; oral: LD50 = 1580 mg/kg                                                                |             |
| 13463-67-7 | 236-675-5 | titanium dioxide                                                                                                                                                                                          | < 1 %       |
|            |           | oral: LD50 = > 2000 mg/kg                                                                                                                                                                                 |             |

### SECTION 4: First aid measures

#### 4.1. Description of first aid measures

##### General information

Change contaminated, saturated clothing. In case of accident or unwellness, seek medical advice immediately (show directions for use or safety data sheet if possible).

##### After inhalation

In case of inhalation of decomposition products, affected person should be moved into fresh air and kept still.

##### After contact with skin

After contact with skin, wash immediately with plenty of water and soap. Seek medical advice immediately.  
Do not wash with: Solvents/Thinner

##### After contact with eyes

After contact with the eyes, rinse with water with the eyelids open for a sufficient length of time, then consult an ophthalmologist immediately.

##### After ingestion

If accidentally swallowed rinse the mouth with plenty of water (only if the person is conscious) and obtain immediate medical attention.  
Do NOT induce vomiting.

#### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms may develop several hours following exposure; medical observation therefore necessary for at least 48 hours.

#### 4.3. Indication of any immediate medical attention and special treatment needed

First Aid, decontamination, treatment of symptoms.

### SECTION 5: Firefighting measures

#### 5.1. Extinguishing media

##### Suitable extinguishing media

- Dry extinguishing powder.
- Carbon dioxide (CO<sub>2</sub>).
- alcohol resistant foam.

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 4 of 16

- Water spray jet

#### Unsuitable extinguishing media

- Full water jet

#### 5.2. Special hazards arising from the substance or mixture

- Carbon monoxide
- Carbon dioxide (CO<sub>2</sub>).
- Nitrogen oxides (NO<sub>x</sub>)

#### 5.3. Advice for firefighters

Special protective equipment for firefighters Protective clothing. In case of fire: Wear self-contained breathing apparatus.

Co-ordinate fire-fighting measures to the fire surroundings.

#### Additional information

Collect contaminated fire extinguishing water separately. Do not allow entering drains or surface water.

### SECTION 6: Accidental release measures

#### 6.1. Personal precautions, protective equipment and emergency procedures

##### General advice

- Provide adequate ventilation.
- Remove persons to safety.
- Safe handling: see section 7
- Personal protection equipment: see section 8

#### 6.2. Environmental precautions

Do not allow to enter into surface water or drains. Cover drains. Adverse environmental effects

#### 6.3. Methods and material for containment and cleaning up

##### For containment

Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents). Treat the recovered material as prescribed in the section on waste disposal.

#### 6.4. Reference to other sections

- Safe handling: see section 7
- Personal protection equipment: see section 8
- Disposal: see section 13

### SECTION 7: Handling and storage

#### 7.1. Precautions for safe handling

##### Advice on safe handling

Personal protection equipment: see section 8 Keep container tightly closed.

##### Advice on protection against fire and explosion

Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.

##### Advice on general occupational hygiene

Work in well-ventilated zones or use proper respiratory protection. Only wear fitting, comfortable and clean protective clothing. Avoid contact with skin, eyes and clothes. Wash hands and face before breaks and after work and take a shower if necessary.

#### 7.2. Conditions for safe storage, including any incompatibilities

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 5 of 16

#### Requirements for storage rooms and vessels

Keep container tightly closed in a cool, well-ventilated place. Keep/Store only in original container.

#### Further information on storage conditions

Keep away from:

- Frost
- Heat
- Humidity

#### 7.3. Specific end use(s)

No information available.

### SECTION 8: Exposure controls/personal protection

#### 8.1. Control parameters

##### Occupational exposure limits

| CAS No     | Substance                         | ppm | mg/m <sup>3</sup> | fib/cm <sup>3</sup> | Category  | Origin |
|------------|-----------------------------------|-----|-------------------|---------------------|-----------|--------|
| 1302-74-5  | Emery, respirable dust            | -   | 4                 |                     | TWA (8 h) |        |
| 409-21-2   | Silicon carbide, respirable dust  | -   | 3                 |                     | TWA (8 h) |        |
| 13463-67-7 | Titanium dioxide, respirable dust | -   | 4                 |                     | TWA (8 h) |        |

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 6 of 16

#### DNEL/DMEL values

| CAS No                   | Substance                                                                                                                                                                                                 | Exposure route | Effect                   | Value |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------------|-------|
| 1675-54-3                | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylen)]bisoxiran                                                                                                                                          |                |                          |       |
| Worker DNEL, long-term   | inhalation                                                                                                                                                                                                | local          | 310 mg/m <sup>3</sup>    |       |
| Consumer DNEL, long-term | inhalation                                                                                                                                                                                                | local          | 55 mg/m <sup>3</sup>     |       |
| Worker DNEL, long-term   | inhalation                                                                                                                                                                                                | systemic       | 4,93 mg/m <sup>3</sup>   |       |
| Worker DNEL, long-term   | dermal                                                                                                                                                                                                    | systemic       | 0,75 mg/kg bw/day        |       |
| Consumer DNEL, long-term | inhalation                                                                                                                                                                                                | systemic       | 0,87 mg/m <sup>3</sup>   |       |
| Consumer DNEL, long-term | dermal                                                                                                                                                                                                    | systemic       | 0,0893 mg/kg bw/day      |       |
| Consumer DNEL, long-term | oral                                                                                                                                                                                                      | systemic       | 0,5 mg/kg bw/day         |       |
|                          | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy}methyl)oxirane |                |                          |       |
| Worker DNEL, long-term   | inhalation                                                                                                                                                                                                | systemic       | 29,39 mg/m <sup>3</sup>  |       |
| Worker DNEL, long-term   | dermal                                                                                                                                                                                                    | systemic       | 104,15 mg/kg bw/day      |       |
| Worker DNEL, long-term   | inhalation                                                                                                                                                                                                | local          | 0,0083 mg/m <sup>3</sup> |       |
| Consumer DNEL, long-term | inhalation                                                                                                                                                                                                | systemic       | 8,7 mg/m <sup>3</sup>    |       |
| Consumer DNEL, long-term | dermal                                                                                                                                                                                                    | systemic       | 62,5 mg/kg bw/day        |       |
| Consumer DNEL, long-term | oral                                                                                                                                                                                                      | systemic       | 6,25 mg/kg bw/day        |       |
| 409-21-2                 | Silicon carbide                                                                                                                                                                                           |                |                          |       |
| Worker DNEL, acute       | inhalation                                                                                                                                                                                                | systemic       | 94 mg/m <sup>3</sup>     |       |
| Consumer DNEL, acute     | inhalation                                                                                                                                                                                                | systemic       | 23 mg/m <sup>3</sup>     |       |
| Consumer DNEL, acute     | dermal                                                                                                                                                                                                    | systemic       | 200 mg/kg bw/day         |       |
| Consumer DNEL, acute     | oral                                                                                                                                                                                                      | systemic       | 13 mg/kg bw/day          |       |
|                          |                                                                                                                                                                                                           |                |                          |       |
| 100-51-6                 | benzyl alcohol                                                                                                                                                                                            |                |                          |       |
| Worker DNEL, long-term   | inhalation                                                                                                                                                                                                | systemic       | 22 mg/m <sup>3</sup>     |       |
| Worker DNEL, acute       | inhalation                                                                                                                                                                                                | systemic       | 110 mg/m <sup>3</sup>    |       |
| Worker DNEL, long-term   | dermal                                                                                                                                                                                                    | systemic       | 8 mg/kg bw/day           |       |
| Worker DNEL, acute       | dermal                                                                                                                                                                                                    | systemic       | 40 mg/kg bw/day          |       |
| Consumer DNEL, long-term | inhalation                                                                                                                                                                                                | systemic       | 5,4 mg/m <sup>3</sup>    |       |
| Consumer DNEL, acute     | inhalation                                                                                                                                                                                                | systemic       | 27 mg/m <sup>3</sup>     |       |
| Consumer DNEL, long-term | dermal                                                                                                                                                                                                    | systemic       | 4 mg/kg bw/day           |       |
| Consumer DNEL, acute     | dermal                                                                                                                                                                                                    | systemic       | 20 mg/kg bw/day          |       |
| Consumer DNEL, long-term | oral                                                                                                                                                                                                      | systemic       | 4 mg/kg bw/day           |       |

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 7 of 16

|                          |                  |          |                      |
|--------------------------|------------------|----------|----------------------|
| Consumer DNEL, acute     | oral             | systemic | 20 mg/kg bw/day      |
|                          |                  |          |                      |
| 13463-67-7               | titanium dioxide |          |                      |
| Worker DNEL, long-term   | inhalation       | local    | 10 mg/m <sup>3</sup> |
| Consumer DNEL, long-term | oral             | systemic | 700 mg/kg bw/day     |

#### PNEC values

| CAS No                                           | Substance                                                                                                                                                                                                 |             |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| Environmental compartment                        |                                                                                                                                                                                                           | Value       |
| 1675-54-3                                        | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylen)]bisoxiran                                                                                                                                          |             |
| Freshwater                                       |                                                                                                                                                                                                           | 0,006 mg/l  |
| Freshwater (intermittent releases)               |                                                                                                                                                                                                           | 0,018 mg/l  |
| Marine water                                     |                                                                                                                                                                                                           | 0,001 mg/l  |
| Freshwater sediment                              |                                                                                                                                                                                                           | 0,341 mg/kg |
| Marine sediment                                  |                                                                                                                                                                                                           | 0,034 mg/kg |
| Secondary poisoning                              |                                                                                                                                                                                                           | 11 mg/kg    |
| Micro-organisms in sewage treatment plants (STP) |                                                                                                                                                                                                           | 10 mg/l     |
| Soil                                             |                                                                                                                                                                                                           | 0,065 mg/kg |
|                                                  | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy}methyl)oxirane |             |
| Freshwater                                       |                                                                                                                                                                                                           | 0,003 mg/l  |
| Freshwater (intermittent releases)               |                                                                                                                                                                                                           | 0,025 mg/l  |
| Marine water                                     |                                                                                                                                                                                                           | 0 mg/l      |
| Freshwater sediment                              |                                                                                                                                                                                                           | 0,294 mg/kg |
| Marine sediment                                  |                                                                                                                                                                                                           | 0,029 mg/kg |
| Micro-organisms in sewage treatment plants (STP) |                                                                                                                                                                                                           | 10 mg/l     |
| Soil                                             |                                                                                                                                                                                                           | 0,237 mg/kg |
| 100-51-6                                         | benzyl alcohol                                                                                                                                                                                            |             |
| Freshwater                                       |                                                                                                                                                                                                           | 1 mg/l      |
| Freshwater (intermittent releases)               |                                                                                                                                                                                                           | 2,3 mg/l    |
| Marine water                                     |                                                                                                                                                                                                           | 0,1 mg/l    |
| Freshwater sediment                              |                                                                                                                                                                                                           | 5,27 mg/kg  |
| Marine sediment                                  |                                                                                                                                                                                                           | 0,527 mg/kg |
| Micro-organisms in sewage treatment plants (STP) |                                                                                                                                                                                                           | 39 mg/l     |
| Soil                                             |                                                                                                                                                                                                           | 0,456 mg/kg |

#### 8.2. Exposure controls

##### Appropriate engineering controls

Provide adequate ventilation as well as local exhaust at critical locations.

##### Individual protection measures, such as personal protective equipment

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 8 of 16

#### Eye/face protection

Suitable eye protection:

Eye glasses with side protection  
goggles

#### Hand protection

Tested protective gloves must be worn: EN ISO 374

NBR (Nitrile rubber), Butyl caoutchouc (butyl rubber)

Thickness of the glove material  $\geq 0,4$  mm

Breakthrough times and swelling properties of the material must be taken into consideration.

For special purposes, it is recommended to check the resistance to chemicals of the protective gloves mentioned above together with the supplier of these gloves.

Wearing time with occasional contact (splashes): max. 480 min. (NBR (Nitrile rubber))

Wearing time with permanent contact 240 - 480 min (NBR (Nitrile rubber))

Observe the wear time limits as specified by the manufacturer.

#### Skin protection

Protective clothing

#### Respiratory protection

Usually no personal respiratory protection necessary.

If technical exhaust or ventilation measures are not possible or insufficient, respiratory protection must be worn.

Combination filtering device A-P3

Self-contained respirator (breathing apparatus)

#### Thermal hazards

No data available

## SECTION 9: Physical and chemical properties

### 9.1. Information on basic physical and chemical properties

|                 |                |
|-----------------|----------------|
| Physical state: | Paste          |
| Colour:         | blue           |
| Odour:          | characteristic |

#### Changes in the physical state

|                               |                   |
|-------------------------------|-------------------|
| Melting point/freezing point: | No data available |
|-------------------------------|-------------------|

|                                                           |                   |
|-----------------------------------------------------------|-------------------|
| Boiling point or initial boiling point and boiling range: | No data available |
|-----------------------------------------------------------|-------------------|

|              |           |
|--------------|-----------|
| Flash point: | $>102$ °C |
|--------------|-----------|

#### Flammability

|               |                   |
|---------------|-------------------|
| Solid/liquid: | No data available |
|---------------|-------------------|

|      |                   |
|------|-------------------|
| Gas: | No data available |
|------|-------------------|

#### Explosive properties

No information available.

|                         |                   |
|-------------------------|-------------------|
| Lower explosion limits: | No data available |
|-------------------------|-------------------|

|                         |                   |
|-------------------------|-------------------|
| Upper explosion limits: | No data available |
|-------------------------|-------------------|

|                            |                   |
|----------------------------|-------------------|
| Auto-ignition temperature: | No data available |
|----------------------------|-------------------|

#### Self-ignition temperature

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 9 of 16

|                                        |                       |
|----------------------------------------|-----------------------|
| Solid:                                 | No data available     |
| Gas:                                   | No data available     |
| Decomposition temperature:             | No data available     |
| pH-Value:                              | No data available     |
| Viscosity / dynamic:                   | No data available     |
| Water solubility:                      | Immiscible            |
| <b>Solubility in other solvents</b>    |                       |
| No information available.              |                       |
| Partition coefficient n-octanol/water: | No data available     |
| Vapour pressure:                       | >1 (air=1) hPa        |
| Density:                               | 2,2 g/cm <sup>3</sup> |
| Relative vapour density:               | No data available     |

#### **9.2. Other information**

##### **Information with regard to physical hazard classes**

Oxidizing properties  
No information available.

##### **Other safety characteristics**

Evaporation rate: No data available

##### **Further Information**

No information available.

### SECTION 10: Stability and reactivity

#### **10.1. Reactivity**

The product is stable under storage at normal ambient temperatures.

#### **10.2. Chemical stability**

Does not decompose when used for intended uses. No known hazardous decomposition products.

#### **10.3. Possibility of hazardous reactions**

Exothermic reaction with: Acid, Oxidising agent

#### **10.4. Conditions to avoid**

Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.

#### **10.5. Incompatible materials**

Acid, Oxidising agent

#### **10.6. Hazardous decomposition products**

Does not decompose when used for intended uses.

### SECTION 11: Toxicological information

#### **11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008**

##### **Acute toxicity**

Based on available data, the classification criteria are not met.

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 10 of 16

#### ATEmix calculated

ATE (oral) 90804,6 mg/kg; ATE (inhalation vapour) 632,18 mg/l; ATE (inhalation dust/mist) 85,033 mg/l

| CAS No     | Chemical name                                                                                                                                                                                             |                    |         |                                          |                                          |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---------|------------------------------------------|------------------------------------------|
|            | Exposure route                                                                                                                                                                                            | Dose               | Species | Source                                   | Method                                   |
| 1675-54-3  | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran                                                                                                                                         |                    |         |                                          |                                          |
|            | oral                                                                                                                                                                                                      | LD50 19800 mg/kg   | Rabbit  | Publication (1958)                       | Rabbits were orally gavaged with test ma |
|            | dermal                                                                                                                                                                                                    | LD50 > 2000 mg/kg  | Rat     | Study report (2007)                      | OECD Guideline 402                       |
|            | inhalation (4 h) vapour                                                                                                                                                                                   | LC50 ca. 24,6 mg/l | Rat     | AMA Arch. Ind. Hyg. Occ. Med. 10: 61-68  | Rats were exposed to 8000 ppm of the tes |
|            | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl}oxirane |                    |         |                                          |                                          |
|            | oral                                                                                                                                                                                                      | LD50 > 5000 mg/kg  | Rat     | Study report (1988)                      | OECD Guideline 401                       |
|            | dermal                                                                                                                                                                                                    | LD50 > 2000 mg/kg  | Rat     | Study report (1988)                      | OECD Guideline 402                       |
| 100-51-6   | benzyl alcohol                                                                                                                                                                                            |                    |         |                                          |                                          |
|            | oral                                                                                                                                                                                                      | LD50 1580 mg/kg    | Mouse   | Cosmet. Toxicol. 11, 1011-1013 (1973) (1 | OECD Guideline 401                       |
|            | dermal                                                                                                                                                                                                    | LD50 > 2000 mg/kg  | Rabbit  | Raw Material Data Handbook, Vol.1:( Orga | EPA OTS 798.1100                         |
|            | inhalation vapour                                                                                                                                                                                         | ATE 11 mg/l        |         |                                          |                                          |
|            | inhalation (4 h) dust/mist                                                                                                                                                                                | LC50 >4,178 mg/l   | Rat     | ECHA                                     | OECD 403                                 |
| 13463-67-7 | titanium dioxide                                                                                                                                                                                          |                    |         |                                          |                                          |
|            | oral                                                                                                                                                                                                      | LD50 > 2000 mg/kg  | Rat     | Study report (1996)                      | OECD Guideline 401                       |

#### Irritation and corrosivity

Causes skin irritation.

Causes serious eye irritation.

#### Sensitising effects

May cause an allergic skin reaction. (2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran;

Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-

[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-

(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl}oxirane)

#### Carcinogenic/mutagenic/toxic effects for reproduction

Based on available data, the classification criteria are not met.

#### STOT-single exposure

Based on available data, the classification criteria are not met.

#### STOT-repeated exposure

Based on available data, the classification criteria are not met.

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 11 of 16

#### Aspiration hazard

Based on available data, the classification criteria are not met.

#### 11.2. Information on other hazards

##### Endocrine disrupting properties

No data available

### SECTION 12: Ecological information

#### 12.1. Toxicity

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 12 of 16

| CAS No     | Chemical name                                                                                                                                                                                          |                  |           |                                 |                                                                                                 |                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------|---------------------------------|-------------------------------------------------------------------------------------------------|--------------------|
|            | Aquatic toxicity                                                                                                                                                                                       | Dose             | [h]   [d] | Species                         | Source                                                                                          | Method             |
| 1675-54-3  | 2,2'-[(1-Methylethyliden)bis(4,1-phenyleneoxymethylene)]bisoxiran                                                                                                                                      |                  |           |                                 |                                                                                                 |                    |
|            | Acute fish toxicity                                                                                                                                                                                    | LC50 3,6 mg/l    | 96 h      | Oncorhynchus mykiss             | Study report (1982)                                                                             | OECD Guideline 203 |
|            | Acute algae toxicity                                                                                                                                                                                   | ErC50 mg/l > 100 | 72 h      | Raphidocelis subcapitata        | Study report (2007)                                                                             | OECD Guideline 201 |
|            | Acute crustacea toxicity                                                                                                                                                                               | EC50 2,8 mg/l    | 48 h      | Daphnia magna                   | REACH Registration Dossier                                                                      | OECD Guideline 202 |
|            | Crustacea toxicity                                                                                                                                                                                     | NOEC 0,3 mg/l    | 21 d      | Daphnia magna                   | REACH Registration Dossier                                                                      | OECD Guideline 211 |
|            | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-([4-(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl)oxirane |                  |           |                                 |                                                                                                 |                    |
|            | Acute fish toxicity                                                                                                                                                                                    | LC50 mg/l > 1000 | 96 h      | Oncorhynchus mykiss             | Study report (1998)                                                                             | OECD Guideline 203 |
|            | Acute algae toxicity                                                                                                                                                                                   | ErC50 mg/l > 1,8 | 72 h      | Raphidocelis subcapitata        | Study report (1993)                                                                             | OECD Guideline 201 |
|            | Acute crustacea toxicity                                                                                                                                                                               | EL50 mg/l > 1000 | 48 h      | Daphnia magna                   | Study report (1998)                                                                             | OECD Guideline 202 |
|            | Crustacea toxicity                                                                                                                                                                                     | NOEC 0,3 mg/l    | 21 d      | Daphnia magna                   | Study report (1984)                                                                             | OECD Guideline 211 |
| 100-51-6   | benzyl alcohol                                                                                                                                                                                         |                  |           |                                 |                                                                                                 |                    |
|            | Acute fish toxicity                                                                                                                                                                                    | LC50 mg/l > 100  | 96 h      | Oryzias latipes                 | Review article or handbook (2009)                                                               | OECD Guideline 203 |
|            | Acute algae toxicity                                                                                                                                                                                   | ErC50 770 mg/l   | 72 h      | Raphidocelis subcapitata        | Review article or handbook (2009)                                                               | OECD Guideline 201 |
|            | Acute crustacea toxicity                                                                                                                                                                               | EC50 230 mg/l    | 48 h      | Daphnia magna                   | Review article or handbook (2009)                                                               | OECD Guideline 202 |
|            | Fish toxicity                                                                                                                                                                                          | NOEC mg/l 48,897 | 30 d      | Fish species                    | <a href="http://epa.gov/oppt/exposure/pubs/episui">http://epa.gov/oppt/exposure/pubs/episui</a> | other: QSAR        |
|            | Algae toxicity                                                                                                                                                                                         | NOEC 51 mg/l     | 3 d       |                                 |                                                                                                 |                    |
|            | Crustacea toxicity                                                                                                                                                                                     | NOEC 51 mg/l     | 21 d      | Daphnia magna                   | Review article or handbook (2009)                                                               | OECD Guideline 211 |
|            | Acute bacteria toxicity                                                                                                                                                                                | (EC50 mg/l) 1385 | 3 h       | activated sludge, domestic      | Study report (1989)                                                                             | OECD Guideline 209 |
| 13463-67-7 | titanium dioxide                                                                                                                                                                                       |                  |           |                                 |                                                                                                 |                    |
|            | Acute fish toxicity                                                                                                                                                                                    | LC50 mg/l > 100  | 96 h      | Carassius auratus               | REACH Registration Dossier                                                                      | OECD Guideline 203 |
|            | Acute algae toxicity                                                                                                                                                                                   | ErC50 mg/l > 50  | 72 h      | Pseudokirchneriella subcapitata | REACH Registration Dossier                                                                      | OECD Guideline 201 |

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 13 of 16

|  |                          |                |          |      |                                                             |                                                |                                                |
|--|--------------------------|----------------|----------|------|-------------------------------------------------------------|------------------------------------------------|------------------------------------------------|
|  | Acute crustacea toxicity | EC50<br>mg/l   | > 100    | 48 h | Artemia salina                                              | REACH<br>Registration<br>Dossier               | OECD Guideline<br>202                          |
|  | Fish toxicity            | NOEC<br>mg/l   | >= 80    | 6 d  | Danio rerio                                                 | REACH<br>Registration<br>Dossier               | OECD TG 210                                    |
|  | Algae toxicity           | NOEC<br>mg/l   | >= 1     | 32 d | Synedra ulna,<br>Scenedesmus<br>quadricauda,<br>Stigeocloni | Environ. Tox.<br>Chem. 31,<br>2414-2422 (2012) | In this study, the<br>authors report the<br>re |
|  | Crustacea toxicity       | NOEC           | > 1 mg/l | 10 d | Chironomus riparius                                         | REACH<br>Registration<br>Dossier               | other: OECD<br>Guideline 219                   |
|  | Acute bacteria toxicity  | (EC50<br>mg/l) | > 1000   | 3 h  | activated sludge,<br>domestic                               | REACH<br>Registration<br>Dossier               | OECD Guideline<br>209                          |

### 12.2. Persistence and degradability

| CAS No    | Chemical name                                                   |          |    |        |
|-----------|-----------------------------------------------------------------|----------|----|--------|
|           | Method                                                          | Value    | d  | Source |
|           | Evaluation                                                      |          |    |        |
| 1675-54-3 | 2,2'-[(1-Methylethyliden)bis(4,1-phenylenoxymethylen)]bisoxiran |          |    |        |
|           | OECD 302B                                                       | 12%      | 28 |        |
|           | Not readily biodegradable (according to OECD criteria)          |          |    |        |
| 100-51-6  | benzyl alcohol                                                  |          |    |        |
|           | OECD 301A/ ISO 7827/ EEC 92/69/V, C.4-A                         | 95 - 97% | 21 |        |
|           | Readily biodegradable (according to OECD criteria).             |          |    |        |

### 12.3. Bioaccumulative potential

#### Partition coefficient n-octanol/water

| CAS No    | Chemical name                                                                                                                                                                                             | Log Pow |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1675-54-3 | 2,2'-[(1-Methylethyliden)bis(4,1-phenylenoxymethylen)]bisoxiran                                                                                                                                           | >= 2,64 |
|           | Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-({2-[4-(oxiran-2-ylmethoxy)benzyl]phenoxy}methyl)oxirane | 2,7     |
| 100-51-6  | benzyl alcohol                                                                                                                                                                                            | 1       |

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 14 of 16

#### BCF

| CAS No     | Chemical name                                                                                                                                                                                                                     | BCF                | Species        | Source                                                  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|----------------|---------------------------------------------------------|
| 1675-54-3  | 2,2'-<br>[(1-Methylethyliden)bis(4,1-phenylenoxy<br>methylen)]bisoxiran                                                                                                                                                           | 31                 |                | Study report (2010)                                     |
|            | Reaction mass of 2,2'-<br>[methylenebis(2,1-phenyleneoxymethyl<br>ene)]bis(oxirane) and 2,2'-<br>[methylenebis(4,1-phenyleneoxymethyl<br>ene)]bis(oxirane) and 2-({2-[4-<br>(oxiran-2-ylmethoxy)benzyl]phenoxy}<br>methyl)oxirane | 150                |                | Other company data (                                    |
| 100-51-6   | benzyl alcohol                                                                                                                                                                                                                    | 1,371              | QSAR model     | <a href="http://epa.gov/oppt/">http://epa.gov/oppt/</a> |
| 13463-67-7 | titanium dioxide                                                                                                                                                                                                                  | > 0,47 - <<br>3,19 | Artemia salina | REACH Registration D                                    |

#### 12.4. Mobility in soil

No information available.

#### 12.5. Results of PBT and vPvB assessment

The substances in the mixture do not meet the PBT/vPvB criteria according to REACH, annex XIII.

#### 12.6. Endocrine disrupting properties

This product does not contain a substance that has endocrine disrupting properties with respect to non-target organisms as no components meets the criteria.

#### 12.7. Other adverse effects

No information available.

### SECTION 13: Disposal considerations

#### 13.1. Waste treatment methods

##### Disposal recommendations

Dispose of waste according to applicable legislation.

##### Contaminated packaging

Non-contaminated packages may be recycled. Dispose of waste according to applicable legislation.

### SECTION 14: Transport information

#### Land transport (ADR/RID)

##### 14.1. UN number or ID number:

No dangerous good in sense of this transport regulation.

##### 14.2. UN proper shipping name:

No dangerous good in sense of this transport regulation.

##### 14.3. Transport hazard class(es):

No dangerous good in sense of this transport regulation.

##### 14.4. Packing group:

No dangerous good in sense of this transport regulation.

#### Inland waterways transport (ADN)

##### 14.1. UN number or ID number:

No dangerous good in sense of this transport regulation.

##### 14.2. UN proper shipping name:

No dangerous good in sense of this transport regulation.

##### 14.3. Transport hazard class(es):

No dangerous good in sense of this transport regulation.

##### 14.4. Packing group:

No dangerous good in sense of this transport regulation.

#### Marine transport (IMDG)

##### 14.1. UN number or ID number:

No dangerous good in sense of this transport regulation.

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 15 of 16

|                                          |                                                          |
|------------------------------------------|----------------------------------------------------------|
| <b>14.2. UN proper shipping name:</b>    | No dangerous good in sense of this transport regulation. |
| <b>14.3. Transport hazard class(es):</b> | No dangerous good in sense of this transport regulation. |
| <b>14.4. Packing group:</b>              | No dangerous good in sense of this transport regulation. |

#### Air transport (ICAO-TI/IATA-DGR)

|                                          |                                                          |
|------------------------------------------|----------------------------------------------------------|
| <b>14.1. UN number or ID number:</b>     | No dangerous good in sense of this transport regulation. |
| <b>14.2. UN proper shipping name:</b>    | No dangerous good in sense of this transport regulation. |
| <b>14.3. Transport hazard class(es):</b> | No dangerous good in sense of this transport regulation. |
| <b>14.4. Packing group:</b>              | No dangerous good in sense of this transport regulation. |

#### 14.5. Environmental hazards

ENVIRONMENTALLY HAZARDOUS: No

#### 14.6. Special precautions for user

No information available.

#### 14.7. Maritime transport in bulk according to IMO instruments

No information available.

### SECTION 15: Regulatory information

#### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

##### EU regulatory information

Restrictions on use (REACH, annex XVII):

Entry 3, Entry 75

##### National regulatory information

Water hazard class (D): 2 - obviously hazardous to water

#### 15.2. Chemical safety assessment

For the following substances of this mixture a chemical safety assessment has been carried out:

2,2'-[(1-Methylethyliden)bis(4,1-phenylenoxymethylen)]bisoxiran

Reaction mass of 2,2'-[methylenebis(2,1-phenyleneoxymethylene)]bis(oxirane) and 2,2'-

[methylenebis(4,1-phenyleneoxymethylene)]bis(oxirane) and 2-{2-[4-

(oxiran-2-ylmethoxy)benzyl]phenoxy)methyl)oxirane

benzyl alcohol

titanium dioxide

### SECTION 16: Other information

#### Changes

This data sheet contains changes from the previous version in section(s): 2,5,6,7,8,9,11,12,15.

#### Abbreviations and acronyms

ADR: Accord européen sur le transport des marchandises dangereuses par Route

(European Agreement concerning the International Carriage of Dangerous Goods by Road)

RID: Règlement international concernant le transport des marchandises dangereuses par chemin de fer

(Regulations Concerning the International Transport of Dangerous Goods by Rail)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

IATA-DGR: Dangerous Goods Regulations by the "International Air Transport Association" (IATA)

ICAO: International Civil Aviation Organization

ICAO-TI: Technical Instructions by the "International Civil Aviation Organization" (ICAO)

## Safety Data Sheet

according to Regulation (EC) No 1907/2006

### ARC BX2(E) Part A, ARC BX5(E) Part A

Revision date: 09.06.2022

Page 16 of 16

CAS: Chemical Abstracts Service (division of the American Chemical Society)  
GHS: Globally Harmonized System of Classification and Labelling of Chemicals  
CLP: Regulation on Classification, Labelling and Packaging of Substances and Mixtures,  
LC50: Lethal concentration, 50 percent  
LD50: Lethal dose, 50 percent  
EC50: Effectice concentration, 50 percent  
DNEL: Derived No Effect Level  
PNEC: Predicted No Effect Concentration  
PBT: Persistent, Bioaccumulative and Toxic  
vPvB: very Persistent and very Bioaccumulative

#### Classification for mixtures and used evaluation method according to Regulation (EC) No 1272/2008 [CLP]

| Classification          | Classification procedure |
|-------------------------|--------------------------|
| Skin Irrit. 2; H315     | Calculation method       |
| Eye Irrit. 2; H319      | Calculation method       |
| Skin Sens. 1; H317      | Calculation method       |
| Aquatic Chronic 3; H412 | Calculation method       |

#### Relevant H and EUH statements (number and full text)

|      |                                                    |
|------|----------------------------------------------------|
| H302 | Harmful if swallowed.                              |
| H315 | Causes skin irritation.                            |
| H317 | May cause an allergic skin reaction.               |
| H319 | Causes serious eye irritation.                     |
| H332 | Harmful if inhaled.                                |
| H351 | Suspected of causing cancer.                       |
| H411 | Toxic to aquatic life with long lasting effects.   |
| H412 | Harmful to aquatic life with long lasting effects. |

*(The data for the hazardous ingredients were taken respectively from the last version of the sub-contractor's safety data sheet.)*